Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int count(int arr[], int left, int right, int num){
    int c = 0;
    for(int i = left; i <= right; i++){
        if(arr[i] == num)
            c++;
    }
    return c;
}

int majority(int arr[], int left, int right){
    if(left == right)
        return arr[left];
    int mid = (left + right) / 2;
    int leftMajor = majority(arr, left, mid);
    int rightMajor = majority(arr, mid + 1, right);
    if(leftMajor == rightMajor)
        return leftMajor;
    int leftCount = count(arr, left, right, leftMajor);
    int rightCount = count(arr, left, right, rightMajor);
    if(leftCount > rightCount)
        return leftMajor;
    else
        return rightMajor;
}
```

```
27
28 int main(){
29     int n;
30     scanf("%d", &n);
31     int arr[n];
32     for(int i = 0; i < n; i++){
33         scanf("%d", &arr[i]);
34     }
35     printf("%d", majority(arr, 0, n - 1));
36     return 0;
37 }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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